

A Unified Framework for Evaluating Systems that Allocate Visibility Algorithmically

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Abstract

Throughout application domains, we now rely extensively on algorithmic systems to engage with ever-expanding datasets of information. Despite their benefits, these systems are often complex (comprising of many intricate tools, e.g., search, recommender systems, prediction models), of unknown structure (due to the lack of accompanying documentation), and having hard-to-predict yet potentially severe downstream consequences (due to the extensive use, systematic enactment of existing errors, and many comprising feedback loops). As such, understanding and evaluating these systems as a whole remains a challenge for both researchers and legislators. To aid ongoing efforts, we first introduce the term *visibility allocation system* (VAS) as a way of referring to (semi-)automated systems deciding which (processed) data to present a human user with. Next, we review typical tools comprising VASs, such as search and recommender systems, and define the associated computational problems they solve. By doing so, VASs can be decomposed into sub-processes and illustrated via data flow diagrams. Moreover, we provide an overview of the different stakeholders and ways they are affected by VASs, arguing for a multi-stakeholder approach to VAS-evaluation. Using forecasting-based recommendations in school choice as a case study, we demonstrate how our framework can support VAS evaluation. We also discuss how our framework can support ongoing AI-legislative efforts to locate obligations, quantify systemic risks, and enable adaptive compliance. Altogether, this work defines VASs and unifies the language for its different components, ultimately reducing the friction of information transfer across research domains and aligning evaluation.

Keywords

AI Systems, Recommender Systems, Information Filtering, Information Retrieval, Information Access, Machine Learning, Data-flow Diagrams, Agent-based Modeling, Algorithmic Fairness

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1 Introduction

Throughout application domains, we have ever-increasing amounts of data at our disposal. To avoid data smog (also referred to as information overload), an algorithmic layer makes a selection for the user [14, 119]. Search engines return lists of articles related to a given query, social media platforms recommend engaging content, online marketplaces provide filters and sorting tools to support satisfactory purchases, while automatic CV assessment tools make a first screening of job candidates. These are just a few examples of technical systems embedded in our society for deciding which *alternatives* (documents, videos, posts, opinions, products, job candidates) to make visible to a given human user. In this work, we will call them *visibility allocation systems* (VASs).

VASs have important societal effects. While some of these effects are positive (e.g., increased access to information, low cost of sharing decision-support systems), others are negative. Extensive reliance on personalized recommendations can lead to convergence in taste [16, 104], echo chambers, and filter bubbles [57, 100]. Machine learning models used in high-stakes decision-making processes can suffer from and even amplify biases because of unsuitable training sets [13, 53] or measurements [54, 72]. More generally, systems of machines influence culture evolution as they generate, select, transmit, and alter cultural artifacts [23]. Due to all their influence on society, both researchers and legislators underline the importance of understanding the long-term effects of such processes [23, 99].

However, evaluating the long-term effects of VASs is challenging for several reasons. First, these systems are complex, consisting of multiple intricate tools (e.g., moderation, search, recommendation systems, prediction models), which are often developed and evaluated in isolation. Second, the lack of transparency of VAS architecture leaves many unknowns, such as the current implementation of comprising tools and interaction between them. Moreover, VASs change frequently, leaving ambiguous which version is being evaluated and whether takeaways still apply to current versions. All these are just a few examples of related challenges.

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Perhaps the biggest challenge in understanding and evaluating VASs is the large number of relevant research fields and methodologies. Computer science focuses on the development of efficient algorithms, Economics on the analysis of strategic user behavior, Psychology on understanding individual human behavior, Sociology on studying the emergence of groups and collective behavior, Law, Ethics, and Philosophy on identifying and deploying desired properties of systems. The emergence of fields such as Human-Computer Interaction, Algorithmic Fairness, and Economics and Computation helps bridge between communities, but cannot, of course, diminish the diversity of relevant expertise. As a result, various methods and procedures offer complementary insight into the study of VASs, e.g., complexity theory, formal verification, algorithmic auditing, experiments, surveys, data analysis, and agent-based modeling.

The goal of our work is to unify the language surrounding (semi)automated systems that decide which (processed) data to present a human user with. By providing a framework for describing and understanding such systems, we aim to lower the friction of information transfer between domains. Ultimately, we show this makes the study of long-term societal effects easier and supports ongoing legislative efforts.

Doing so leads to several contributions. Namely, we:

- (1) Define the visibility allocation problem and systems.
- (2) Survey the most frequently used tools within VASs (e.g., search, filter, recommendation, moderation). We provide compatible definitions for them, effectively decomposing VASs into sub-processes for a more granular evaluation.
- (3) Demonstrate how VASs can be represented by data-flow diagrams (DFDs) and their comprising tools as processes within DFDs. Logs can document systemic changes and learnings. Such high-level system representations promote better understanding and increase transparency.
- (4) Review both tool-specific and long-term societal effects relevant to studying VASs. Executable versions of DFDs help connect design choices to their long-term effects.
- (5) Using forecasting-based recommendations in the context of school choice as a case study, demonstrate how to operationalize our framework and how this framework aids in the evaluation of long-term effects of VASs.

2 Brief Overview of Related Work

Other (semi)-automated systems. While already existing system definitions bear similarities to the *visibility allocation systems* (VASs), they are overall distinct. *Information access systems* also aim to present users with items that satisfy their information need [45], and encompass both *information retrieval* (IR) and *information filtering* (IF) [17, 66]. Still, they do not include, e.g. machine learning models trained to make predictions based on offline interaction data. While *decision support systems* include such models [22], they do not cover other processes that influence both individuals and society, such as ad auctions [7]. This is why prior work resorts to local, ad-hoc terms like ‘the algorithm’ [64], semi-autonomous algorithmic technologies [104], systems for content processing and propagation [99]. To cover this gap, we introduce VAS for referring to all technical systems that decide which alternatives are visible

to users. Similar to this prior work, we define VASs based on the computational problem they address [45]; additionally, we provide definitions for the problems associated with the comprising tools.

Societal impact of VASs. Most studies focus on the societal impact of either specific tools or specific instances of platform design. For example, machine learning models can learn biased representations from training data [30], leading to algorithmic biases and the reinforcement of existing societal inequalities [102], effects often amplified through feedback loops [87, 103]. At a platform level, studies on social media show they can contribute to the spread of misinformation [130, 131], create echo chambers [35, 41], and lead to polarization [55]. Platform changes can reshape user behavior: e.g., the introduction of the feature ‘People You May Know’ on Facebook led to a drastic increase in triangles and new connections [135]. Our VAS framework supports documenting such systemic changes and learnings, as well as connecting specific design choices to their societal impact and transferring learnings across application domains.

Relevant research fields and methodologies. More work aiming to integrate multidisciplinary perspectives is needed to ensure stakeholder-aligned system design [83, 84]. Computer science provides the technical foundation through algorithm development and audits, focusing on system architecture and optimization [93, 112]. Yet, it does not give a complete picture. Fairness, transparency, and accountability in AI, for example, need to be grounded in the broader legal, social, and ethical contexts [33, 124]. Mitigating structural inequalities in algorithmic decision-making requires additional insight from social sciences [56]. Moreover, policy and system design changes can alter individual behavior. To this end, economics uses game theory, mechanism design, and agent-based models to investigate changes in incentives that could lead to unwanted strategic behavior [129], while behavioral economics and psychology examine bounded rationality and contextual influences on individual decision-making [10, 61, 77]. By providing harmonized terminology with shared representations and definitions, we aim to facilitate cross-disciplinary understanding. Through structured diagrams and tables, our framework supports a unified tracking of systemic changes, new algorithmic developments, and findings from data analysis and experiments carried out on particular VASs or demographics; this helps to quickly integrate and transfer knowledge across domains.

Legislation. As both the tools comprising VASs and specific implementations of VASs are known to have important societal consequences, they are also the subject of current regulatory efforts. On AI Systems alone, there are several key pieces of legislation: EU’s Artificial Intelligence Act [50], Digital Services Act [49], and the General Data Protection Regulation [48]. Additionally, UNESCO’s Recommendation on the Ethics of AI [126] as well as US Executive Order 14110 [127] (revoked in January 2025) have served as influential normative reference points. These aim to provide harmonised rules regarding the development and deployment of AI systems, promote transparency (a shared understanding among market operators, public awareness), identify and regulate high-risk systems. Our framework intends to support these efforts by providing a

harmonized language, suggesting structured diagrams with accompanying documentation, and using these to more easily integrate learnings (e.g., into executable models) and identify knowledge gaps. We demonstrate how to use this framework to evaluate high-risk systems in one of the critical sectors, education [49].

3 Fundamentals of Visibility Allocation Systems

We call a (semi-)automated system deciding which (processed) data to present to a given human user a visibility allocation system (VAS). VASs, thus, provide a way of deciding which user sees what, i.e. a way of allocating visibility over a set of alternatives. More precisely, we define the visibility allocation problem as follows:

Given a dataset of alternatives and inputs from stakeholders, present a user with a structured subset (from the space) of alternatives satisfying the needs and requirements of the stakeholders.

Above, *structured subset* refers to a subset of alternatives arranged for display in a Graphical User Interface; this rendering aspect is important to include as it can influence both accuracy and fairness [15, 31, 45]. Stakeholders can provide *input* both explicitly (e.g., likes, ratings, filter criteria) and implicitly (e.g., via clicks, watch time) [117]. We opt for *alternatives* instead of *items* for two main reasons. First, to emphasize the term can not only refer to items in the classical sense (e.g., products, documents, news, or videos), but also to people (e.g., content creators, job applicants, students, or refugees). Second, to also capture alternatives in a continuous space, such as opinions. Moreover, we say the output can be *from the space of alternatives* to account for more recent tools such as summarization techniques and large language models which can generate new texts that fall outside the dataset of existing documents.

Visibility Allocation Tools

VASs are composed of a variety of algorithmic tools, which share a common operational logic: they take in a set of alternatives, apply some transformation or evaluation, and output a refined subset or reordered version of those alternatives. Each tool can be understood in terms of three key dimensions: input, design choices, and output. Input refers to the data a tool processes, typically a set of alternatives, and any user-specified parameters, such as filters, queries, or sorting preferences. Design choices are the internal mechanisms or policies that guide how inputs are handled. These include algorithms, evaluators, or rules, which vary depending on the tool's function. Output is the result produced after processing, which may be, e.g., a filtered subset, a ranked list, or a classification decision.

Filter tools select a subset of alternatives based on specified conditions. The input typically includes a set of alternatives and, optionally, user-specified filters or classifiers. The design of a filter tool involves defining evaluators which are functions that assign a binary value to each alternative, determining whether it meets the filtering criteria. The output is a reduced set of alternatives that satisfy the selected evaluators, effectively narrowing down what is visible to the user [89]. Thus, the filter problem is:

Given a dataset of alternatives, a set of evaluators, and a user's set of evaluators, present the maximal subset of alternatives matching the user's evaluators.

Filtering tools play an important role across various contexts: for example they are used to design word filters for moderating comments on social media [74], to control youth access to online content [95, 111], to allow users to systematically filter out alternatives based on predefined criteria on online dating platforms [69].

Sort tools reorder alternatives based on a chosen criterion [80]. They take as input a set of alternatives, often along with a user-specified sorting preference such as relevance, date, or popularity. The design involves selecting sorting criteria and a sorting algorithm. The output is a permutation of the alternatives, arranged so that their values on the chosen criterion follow a defined order (e.g., descending relevance). Formally, sort tools solve the following problem:

Given a dataset of alternatives, a key function assigning a value to every alternative, return a permutation of the alternatives ordered by their respective values.

Sorting tools allow users to change the order based on their preferences or needs. For example, online marketplaces allow sorting by price, ratings, or newest arrivals [34]. Social media and news platforms offer sorting by chronological order or trending [120]. By providing specific sorting choices or setting a default one, platforms exert considerable influence over which information users encounter first, thereby shaping user attention and impacting user decisions [47, 59, 114].

We regard filter and sort as foundational because other tools can be viewed as functional compositions of them, often extended with scoring functions. Search, for instance, consists of retrieval (that **filters**, e.g., documents containing at least one of the query's terms) and ranking (that **sorts** retrieved documents based on their fit with the query, as estimated by a scoring function) [98]. Other examples are recommender systems, moderation, and forecasting tools. Their added complexity brings important differences, such as more intricate design choices and societal effects.

Search tools help users find information by returning content in response to a query [80]. Input includes a user-specified search criterion, usually a query, and a set of alternatives to search through. Design choices determine how the tool compares the query to each alternative in the dataset. The output is a subset of alternatives that best match the search criterion, typically presented in order of estimated relevance. Search tools address the following problem:

Given a dataset of alternatives, a user query, return a subset of these alternatives scored by their fitness with the query.

Search tools play a central role in VASs by helping users find information. They are integrated into on-site site search bars, academic databases, and most prominently, web search engines. Among these, Google Search has become the dominant gateway to online information - influencing how and what news, knowledge, products and services billions of users access, thereby shaping public understanding, attention, and even political discourse [20, 46, 85, 105]. These tools increasingly satisfy information needs directly within the search engine results page, eliminating the need to navigate to external websites and underscoring their central role in shaping user access to information [60].

Recommender Systems (RSs) are personalized information filtering systems: they collect information on the preferences of

users to improve the user's experience by presenting content based on their individual tastes [4]. The input consists of a set of users with their associated attributes (e.g., past interactions, demographics, or behavioral patterns), user needs (either inferred or explicitly expressed), and a set of alternatives (usually referred to as items). Designers choose a recommendation algorithm (e.g., collaborative filtering), thus effectively weighting the criteria used for recommendations. The output is a score for the fitness of each item for a given user. Based on these scores, the system produces a personalized ranked list of recommendations. Formally, RSs address the following problem:

Given datasets of alternatives, user information, and an (implicit or explicit) user need, return a subset of alternatives predicted to best satisfy the user need.

RSs are used to reduce users' effort and time searching for relevant content in various domains like tourism, e-learning, e-commerce, movies, and news [116]. They personalize and control which items are visible, making them critical for VASs.

Moderation tools are typically used on online platforms to remove content or exclude users, and can be manual, similar to filtering, or automated [65]. Automated content moderation aims to identify and regulate content that might be inappropriate or doesn't align with the platform's guidelines [62]. By applying algorithms to detect hate speech, misinformation, harmful or hateful content, these systems can adjust the visibility of such content and thus modify or suppress it. The input consists of a set of alternatives, generated by contributors of the platforms. The design is centered around a moderation function, often based on a policy that defines acceptable content. The moderation function assesses whether an alternative aligns with a given set of content policies. Depending on the outcome of this function, the tool may take actions such as removing the content, or flagging it as sensible.

Given a dataset of alternatives, a set of policies, a set of possible moderation actions, determine whether each alternative aligns with the policies and choose the appropriate action.

Online platforms and websites host an increasing volume of user-generated content, which can lead to an increasing volume of harmful, false, and dangerous content [21, 78]. Consistent and transparent moderation that can be enforced by different stakeholders is thus a key component of VASs.

Forecasting tools, widely used for decision-making and planning, support visibility decisions by making predictions about the future based on historical knowledge [40, 109]. Their input includes past and present data, and sometimes partial information about the near future (e.g., planned events or expected signals). Design choices vary widely and may include statistical models, machine learning algorithms, or domain-specific rules that account for trends, seasonality, uncertainty, or risk. The output consists of predicted future values, often accompanied by risk assessments, confidence scores or probability intervals, offering guidance on likely developments that may inform visibility decisions downstream.

Given a dataset of historical observations (i.e., pairings of features to outcomes) and a set of new inputs (i.e., features), return predicted future observations for each instance in this set.

Petropoulos et al. [109] present a wide range of forecasting methods and practical examples of their applications. Forecasting tools have been extensively used for resource planning and allocation in, e.g., school admissions, refugee assignment, or medical residency matching [11, 90, 132]. In such domains, decision makers can scrutinize alternatives based on their predicted chances of achieving a positive outcome. This can ultimately have a great impact on the individuals about which decisions are being made (e.g., students, refugees). For example, biased datasets or algorithms can disproportionately affect minorities or disadvantaged groups and reinforce inequalities within society [96]. Moreover, as we show in the final case study of this work, these predictions impact the very population they aim to predict; this leads to a feedback loop known as performativity, where predictions can have a causal influence on the decisions of agents that build the datasets of observations [67, 71, 108].

Additional tools. While we consider the tools presented above to be the primarily used ones, the list is far from exhaustive. Ad auctions [109] and query suggestions [101] are two other examples of tools that affect what is visible to users. Moreover, increasingly many people use large language models (LLMs) to inform themselves on a variety of topics [51, 82]. While users rely on agentic discovery, developers include LLMs in visibility allocation pipelines (e.g., through generative retrieval). This recent large-scale adoption of LLMs demonstrates how new visibility allocation tools continue to emerge and evolve. This makes providing a full list of tools both unfeasible and not useful in the long run. Instead, our hope is the tools presented above can serve as examples of how to formulate other visibility allocation tools as data transformers and visibility selectors. As we will demonstrate in the upcoming sections, viewing them as such can facilitate an analysis of the system as a whole, connect design choices with long-term effects, and transfer learnings about these effects between the different tools. Ultimately, we believe *visibility allocation system* is a needed terminology to comprise the tools commonly used nowadays for accessing information, one which we hope will support the research in the field in a similar way as *information retrieval* (IR) and *information filtering* (IF) [17, 66] did for their respective sets of tools.

4 Illustrating and Evaluating VASs

The main difficulty in analyzing VASs is connecting tool-level design choices with long-term effects. On the one hand, stakeholders are primarily interested in the long-term effects of VASs: companies in their future revenues, end-users in their over-time satisfaction with the shown alternatives, society and legislators in how they affect societal outcomes. However, developing VASs ultimately requires decisions about the design of each particular tool, which are difficult to connect with the resulting long-term effects of a VAS as a whole. The multitude of stakeholders with often non-aligning goals adds to the complexity of analyzing alternative designs of VASs and choosing between them.

To this end, this section introduces and adapts data-flow diagrams (DFDs) for representing VASs within the social context they are deployed. It also reviews commonly used metrics, existing challenges in evaluating VAS-based sociotechnical systems, and how our integrated framework of tool definitions and VAS representation supports connecting design choices to long-term effects.

Data Flow Diagrams for Sociotechnical Systems

Data flow diagrams (DFDs) are one of the fundamental structured methodologies for the analysis and design of information systems [42, 121, 133]. They are graphical representations showing the flow of data throughout a system. Processes (represented by circles) take incoming data and transform it, thus producing new data. Besides other processes, the data can originate or go to data-stores (open-ended rectangles) or external entities, such as users and producers (rectangles). Arrows capture the direction of data flow and can be annotated with the data being transferred. Figure 1 shows these elements for a social media platform VAS comprising of moderation, search, and a recommender system.

To adapt DFDs for the study of sociotechnical systems, we make two changes. First, we argue for considering the comprising tools (reviewed in the previous section) as the main processes. This departs from the original focus of DFDs on modeling process structures for business, many of which have limited societal effects [52], e.g., *recording payment*, *producing payment confirmation*, and *account creation* [29, 42]. Second, external entities (stakeholders) are no longer simple sources or sinks that are outside the domain of our study [42], but could be an active part of processes. For example, when a new YouTube video (item) is created, the content creator (producer) might interview other people, discuss news, or events in a specific cultural context. Thus, the creation of videos becomes a process in which both the producer and society take part. We represent such relationships by dotted arrows. The appendix further discusses the choice of DFDs and the changes we made.

VAS Effects on Different Stakeholders

A key role of dataflow diagrams (DFDs) is to represent the stakeholders involved within a sociotechnical system together with their role and interactions within this system. In fact, context diagrams (DFDs at the highest level of abstraction) just illustrate the VAS and the stakeholders interacting with it [39]. Figure 2 provides an example of a context diagram for the VAS introduced earlier. It succinctly shows the main stakeholders, and how they influence and are influenced by the VAS. Some of these influences are direct (e.g., consumers directly provide queries to the VAS) while others are indirect (e.g., society changes based on the updates in preferences and beliefs of individuals). Naturally, distinct stakeholders are interested in different aspects affected by VASs, which could be at odds. As such, a complete analysis of VASs accounts for metrics of interest to the various stakeholders.

Users. Users are the primary consumers of VASs. They engage with ranked or filtered content to satisfy information needs, make decisions, or participate in social discourse. Their experience of the system is typically guided by personalized recommendations, search results, or curated feeds. Users may have limited awareness and control over the mechanisms producing what they experience. They provide, however, input (e.g., via clicks, queries, and ratings) which is fed back into the system, ultimately influencing future VAS outputs [47, 113]. Users may prioritize metrics related to perceived service quality and cost-effectiveness, such as recommendation relevance and user satisfaction scores.

Producers. Producers include those who design, implement, and supply content to VASs, such as developers and content creators. While developers focus on optimizing algorithmic components, content producers rely on visibility to reach audiences, drive engagement, and generate revenue. Although producers can shape visibility allocation through strategic behavior, they often remain constrained to platform policies and shifting algorithmic logic [24, 26, 59]. From a producer perspective, metrics may focus on engagement metrics like clicks, conversion rates, and overall visibility of their content, which directly impact their revenue and reach.

Platforms. Platforms serve as the infrastructure hosting VASs, connecting producers and users and facilitating their interactions within the system. They maintain the technical infrastructure, design the visibility allocation logic, and often govern behavior through moderation and curation, especially in centralized platforms. Moreover, platforms have an influence on who gets seen and also have a stake in being seen themselves, leading to potential conflict of interests [24, 26, 110]. Platforms emphasize system performance metrics such as accuracy, F1 scores, user satisfaction, usability and profitability, which are crucial for optimizing operational efficiency and market competitiveness.

Additional Stakeholders. Additional stakeholders include the general public, regulators, and the natural environment. While not directly involved in interactions with VASs, these actors are affected by their broader societal and ecological impacts. Systemic effects of VASs on additional stakeholder include the polarization of public discourse, unequal access to critical information (e.g., health or political content), and the spread of misinformation. VASs may also reinforce societal biases or contribute to cultural homogenization by privileging dominant perspectives [18, 101, 107, 125]. Moreover, the environmental footprint of operating large-scale algorithmic systems, from energy-intensive data centers to device obsolescence, affects planetary sustainability [19, 37, 88]. These externalities highlight the need for evaluation frameworks that go beyond user- and platform-centric metrics to account for societal risks, environmental costs, and broader patterns of knowledge diffusion and market dynamics.

From Design-choices to Long-term Evaluation

For operational reasons, most of the system evaluation is carried out at a tool level. For instance, when filtering out undesirable comments on social media platforms, developers and researchers measure content creator satisfaction and accuracy [74, 122]. In sorting, we measure the performance of algorithms through time complexity and stability [5]. Search engineers, on the other hand, might be concerned with search satisfaction metrics like precision or completeness, document relevance and clicks [32]. Recommendation systems use various performance metrics like precision and recall [1, 116].

While these metrics are vital for evaluating the accuracy and reliability of tools within VASs, they fall short of capturing the time dimension and system evolution. For example, well-received but biased filter criteria on online dating platforms could lead to racial homogeneity, ultimately exacerbating income inequalities [69]. Surveys and experiments help identify such issues, leading to social

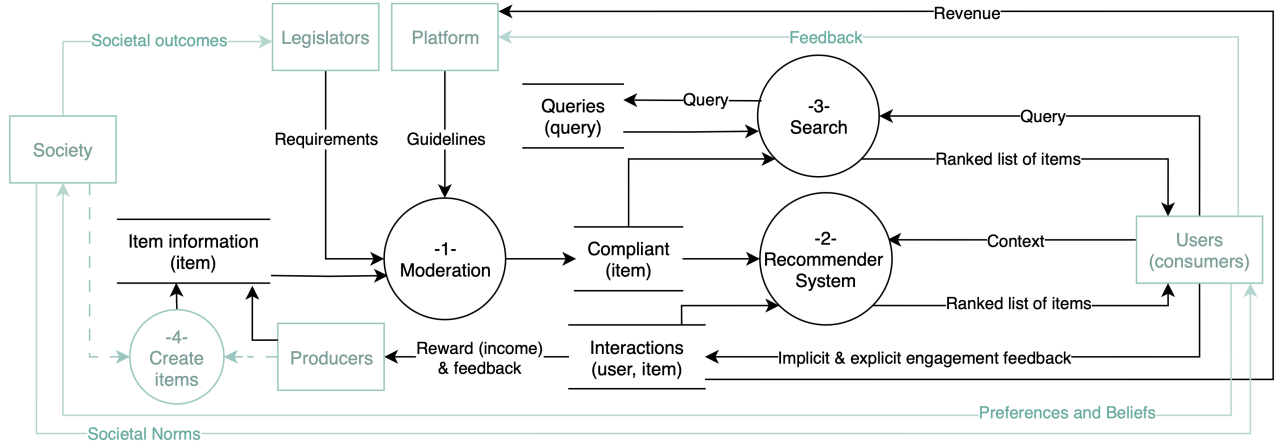


Figure 1: Data-flow diagram for a platform with user-generated content. Dark (black) components are part of the Visibility Allocation System (VAS) with three tools: moderation, recommender system, and search. Light (green) components are external to the VAS. The diagram includes: stakeholders (rectangles), processes (circles), information added to the dataset (open-ended rectangles) indexed by key (annotated in brackets), data flows (full arrows), and participation in a process (dotted arrows).

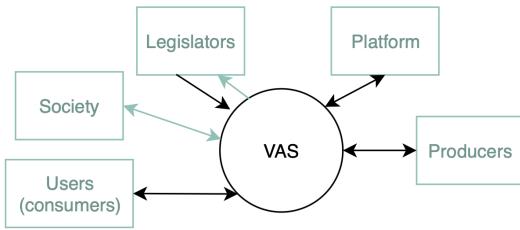


Figure 2: The context diagram showing the VAS for user-generated content in Figure 1 at the highest level of abstraction (level 0). Dark (black) arrows show direct impact, i.e., via direct information transfer. Light (green) arrows show indirect impact, i.e., via VAS-external influence.

theories which can ultimately be tested with agent-based models and simulations [9, 71, 134]. More generally, many societal relevant metrics, like fairness, are not static and require simulations for connecting low-level design choices with higher-level effects [38].

To further support interdisciplinary analysis connecting design choices to long-term effects, we suggest accompanying DFDs by logs that document chronological design changes of VASs and findings about specific processes. Doing so can help in several ways. For example, it can help auditing studies shortlist alternative features they could vary and be precise about which VAS design they investigate. It also supports agent-based models overview and integrate the most recent findings, thus producing more realistic and insightful models. Notably, it aligns with ongoing legislative efforts aiming to provide documentation standards for continuous monitoring and increased transparency [50].

5 Example: VAS for Matching Markets

Using school choice as an example, we demonstrate how to construct the data-flow diagram, why such a diagram is useful for

documenting changes in the system as well as for summarising existing knowledge, and show how it can reveal important societal effects of automation. Finally, we discuss how similar principles and learnings apply to other application domains using data-driven allocations.

Constructing the Diagram

In large cities, many new middle-school students arrive every year to be assigned to one of the several existing high schools. This is a difficult task that requires taking into account not only the preferences of students but also several other signals and constraints (e.g., results of students, commute time, siblings already studying at one high school). This is why many governments use a centralized matching mechanism, i.e., students submit their preference (ranking of high schools), and the mechanism returns an allocation (matching each student to one of the high schools) [2, 36]. Because it is difficult for students to decide which high school would be best for them, prior efforts suggested leveraging past data, e.g., to predict students' development at different schools based on prior students' evolution [132].

We can depict this sociotechnical system using a data-flow diagram (Figure 3). This system has two main stakeholders, namely schools and students. The data store keeps track of relevant information about the students, including historical outcomes (e.g., final exam/SAT results) of students who attended courses at different high schools. The VAS consists of three processes: (1a) an information elicitation aimed to gather students' information, (1b) an ML algorithm that, given a dataset of past outcomes of students who attended courses at different high schools, builds a model that predicts outcomes for all new student-highschool pairs, and (1c) a way of presenting the alternative schools to students based on these predictions. There are also two processes external to the VAS: (2) a matching mechanism that makes an allocation for a given set of preferences and (3) the student-school interactions which produce new data.

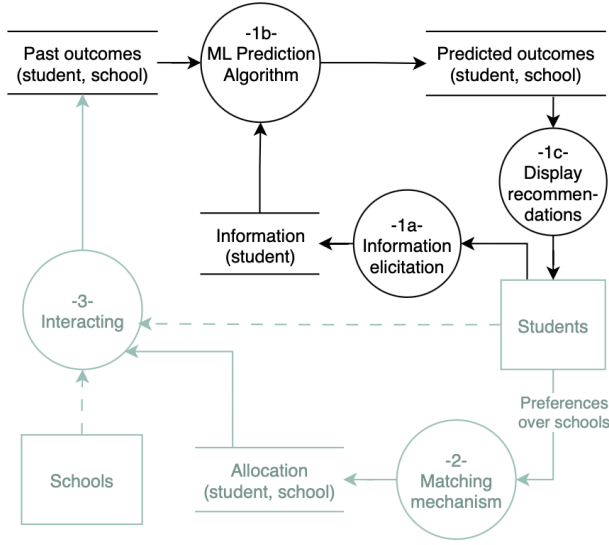


Figure 3: A data-flow diagram for school choice. Dark (black) components are part of the VAS. Light (green) components are external to the VAS. The diagram includes: stakeholders (rectangles), processes (circles), information added to the dataset (open-ended rectangle) indexed by key (annotated in brackets), data flows (full arrows), and participation in a process (dotted arrows).

Supporting System Evaluation

Identifying impacted stakeholders. A first benefit from using a DFD is identifying the stakeholders and how they interact with the VAS. In our case, these stakeholders are students and schools. As such, the dataset stores important information about these agents at different points in the system, e.g., past and predicted outcomes. Keys (i.e., unique identifiers for each agent) help track results across time and processes. Ultimately, this supports a multistakeholder evaluation of the sociotechnical system [1].

Studying the processes in isolation. The first layer of analysis is to study processes in isolation (i.e., by analyzing the way they transform data adjacent in the diagram), which is the focus of most prior work. For instance, we know predictions (process 1b) vary in their levels of accuracy [92] and bias [53]. The design and reputation of the VAS (process 1) can affect the trust of users in the system, and, as a result, the impact of its recommendations on the final preferences of students [44]. Specifically for the school-choice VAS, surveys show most parents were very satisfied with the provided support [132], thus potentially suggesting high levels of trust in the system. The choice of matching mechanisms (process 2) can lead to different properties of the resulting allocation, such as strategy-proofness and stability. Table 1 summarizes these learnings. Next, we show how these data-based studies support a holistic analysis of the sociotechnical system.

Analyzing sociotechnical systems as a whole. A direct analysis of the DFD can reveal new feedback loops, i.e., cycles in the graph

Pr.	Learning
1b	One of the main goals is to improve the accuracy of predictions [92]. Algorithms may have biases; e.g., disparate impact (different rates of outcomes depending on user demographics) [45, 53].
1	Algorithmic systems face different levels of users' trust and algorithmic aversion/appreciation (i.e., users' reluctance/willingness to rely on algorithmic as opposed to human advice) [27, 44, 76]. For school choice, surveys show parents were satisfied with the provided support [132].
2	Prior work studied different matching mechanisms (e.g., serial dictatorship, Boston, Deferred Acceptance). Only some are strategy-proof (i.e., students achieve the best outcome when providing their true preferences) [3].

Table 1: Documents learnings from prior work on the processes (Pr.) depicted in Figure 3.

showing how datasets both influence and are influenced by processes. Figure 3 shows that past outcomes influence recommendations (predicted outcomes), which affect students' preferences and allocations, which impact the next round of outcomes. The diagram shows that both students and schools take part. Ionescu et al. [70] discovered this feedback loop by representing the entire sociotechnical system (without a formal method of doing so) only about 15 years after Wilson et al. [132] proposed the VAS for school choice. Results show this feedback loop can incentivize schools to act strategically, and exacerbate inequalities in the student population. Keeping updated versions of the DFDs could support finding such issues faster.

Evaluating the effects of different design choices. The information in Table 1 can be brought together into one model in order to reason about the entire system. For example, by abstracting an economic model, prior work proved that schools have an incentive to interact strategically in order to trigger different predictions and allocations¹ [70]. An alternative is to create an agent-based model with all the information and implement each process using a programming language, thus building an executable version of the DFD in Figure 3. Results provide an insight into the long-term outcomes (e.g., student inequalities, magnitude of utility gains by interacting strategically) under different design choices and market characteristics (e.g., matching mechanisms, accuracy of the prediction model, market competition). This demonstrates that using modeling to make DFDs executable can test the long-term effects of sociotechnical systems under alternative system designs.

Guiding future research questions. By creating and analyzing DFDs we can identify gaps for future research to fill. In school choice, the diagram showed the presence of the interaction-allocation

¹I.e., high schools which do not want to be matched to under-performing students (e.g., because such students need more costly support for performing well) can interact strategically (by not providing additional support for such students). This signals the VAS not to recommend this school to future under-performing students which leads to fewer matches between the school and such students in the future. For the under-performing students, however, such strategic interactions immediately lead to worse results.

feedback loop. A review of prior work on the comprising processes (Table 1) reveals little is known about strategic interactions (process 3). While these are often optimal in theory [70], future work can use surveys (e.g., as done for strategic reporting in course allocations [25]) and experiments (e.g., via games [28]) to assess the extent to which agents would find and use optimal strategic interactions. If the results confirm that strategic interacting is also a problem in practice, future work can investigate how to best redesign the system. As underlined by Gilbert and Troitzsch [58], modeling and simulations can thus highlight important directions for future research.

Documenting System Evolution

We can use DFDs and chronological reference tables to record changes in sociotechnical systems and their VASs. Table 2, for example, documents the changes for the school choice system in Boston. The matching mechanism (process 2) was the first process to automate, and then update its algorithm on three separate occasions [2, 3, 91]. Moreover, Wilson et al. [132] suggested adding a decision support system for students based on predictions of their development in different schools (process 1) as an alternative to other offline processes (e.g., advice from counselors and anecdotic examples based on other students' experiences). Based on Table 2, one can easily reconstruct the data-flow diagram in Figure 3 depicting the system at different times.

Accompanying DFDs with chronological reference tables documenting VAS changes is important for several reasons. First, the literature introducing changes also motivates them. Thus, this literature provides important learnings that could be transferred to new contexts (e.g., countries) implementing similar VASs. For example, Boston Public Schools (BPS) changed the mechanism from Boston to Deferred Acceptance (DA) because the first was not strategy-proof, i.e., students could be strategic and misreport their preferences. Policymakers argued that non strategy-proof mechanisms are unfair as they "provide an advantage to families who have the time, resources and knowledge to conduct the necessary research" [3]. This motivated the later proposal for using DA in the Chile school choice setting [36]. Second, such a concise and transparent documentation of system design helps understand on which version of the systems prior audits, data analysis, or surveys were conducted. This is key for reasoning on whether results still apply or a new analysis is needed. Third, Table 2 shows the algorithms currently in use, making them easy to incorporate into more realistic agent-based models.

Transfer Learnings to Other Applications

Outcome-driven allocations are also used in other matching markets such as labour market [106], refugee assignment [12], and course allocation [81]. There are, however, differences. For example, while the school choice VAS presents recommendations to students, in refugee assignment, it solves a constrained optimization problem before suggesting an allocation to placement officers [12]. Despite this difference, the two settings are both susceptible to adversarial interaction attacks [70], demonstrating that learnings about one VAS could transfer between domains. We, thus, believe harmonized

Year	Change
1987	Decision of BPS to adopt a choice-based assignment plan (process 2).
1999	BPS adopt the Boston Mechanism (for process 2). [2]
2001	The No Child Left Behind Act of 2001 allows students at low-performing schools (failing assessment goals in two consecutive years) to change institutions.
2005	Boston's School Committee changed the matching mechanism in use (algorithm for process 2, from Boston to Deferred Acceptance) to remove the need for strategic reporting by parents and improve fairness. [3]
2008	Introduces process 1 (prediction alg.): SmartChoice is developed and deployed for a focus group in Charlotte-Mecklenburg [132]
2013	Boston Public Schools changes process 2 to the Home-Based Student Assignment Policy. [91]

Table 2: Timeline documenting changes in school choice; for simplicity, it focuses on Boston Public Schools (BPS).

representations and documentations of VAS designs across applications can facilitate fast transfers of learnings.

6 Discussion

Managing large system documentation. As with any modeling technique, there is a natural tradeoff between simplicity and completeness. Out of the box, data-flow diagrams (DFDs) only partially manage this by having different levels of abstraction. For example, DeMarco [43] notes in a later edition of his original text promoting DFDs that, although project teams still regard them as useful, these DFDs can take up an entire wall of a war room and become hard to maintain. However, recent collaborative tools such as JGraph Ltd [73], Miro [94], SmartDraw Software, LLC [123] can support more tractable ways to work on, maintain, and document DFDs. The resulting diagrams could be made interactive (e.g., by clicking on a process to expand it into a lower-level of abstraction DFD). Moreover, Cheema et al. [29] developed an interface using natural-language processing and rule-based algorithms to draw data flow diagrams. All these tools can help maintain an overview and detailed documentation about large sociotechnical systems, thus improving their transparency.

Beyond data-flow diagrams. While DFDs are one of the cornerstone methodologies in System Analysis and Design, this field now also uses Object-oriented methodologies and the Unified Model Language (UML) [121]. The latter includes for example class diagrams, which provide many additional details that can support system implementation into an executable version. Despite the advantages of these more recent approaches, we opted for DFDs here for several reasons. First, focusing on how data is transformed by tools and processes facilitates evaluations. For example, disparate impact is measured directly on data sets [53]. Moreover, Alshareef et al. [8] provide an algorithm for transforming regular DFDs into Privacy-Aware DFDs by inserting privacy checks. Second, DFDs require less notation, thus making them more accessible across expertise. This aspect is crucial for facilitating the interdisciplinary work needed for the study of sociotechnical systems [56]. Third, requiring companies to provide the detailed specifications typical of UML diagrams

can be impractical, as it may expose proprietary implementation details and undermine competitive advantage [75, 115]. Finally, DFDs can be integrated with other diagram representations to provide additional insight [6, 79].

Towards an integrated simulation environment. We also envision this work as a first step toward building an integrated simulation environment for VAS-based sociotechnical systems. Many research fields extensively rely on such environments to analyze, test, and compare alternative settings in a fast, risk-free, and reproducible manner. Examples include OMNeT++, used for communication networks, multiprocessors, and other distributed or parallel systems [128], NEURON - for biological models of neuronal networks [68], and MASON - a toolkit for general multi-agent systems [86]. By defining the visibility allocation problem, we also specified the scope of such a simulation environment. Through decomposing this problem into tool-specific sub-problems, we provide a unifying standard for the inputs and outputs of the tools; as a result, we can chain different implementations of these tools to test alternative VAS designs. These specifications can also serve as guidelines for developing new or adapting existing implementations of processes. As a result, researchers and developers in different fields can independently contribute with their own implementation (e.g., for recommender systems [63] or human behavior in response to VAS-outputs [97, 118]) toward the final analysis. Data-flow diagrams comprehensively describe the alternative designs being evaluated. Altogether, we believe that by continuing this line of work with building such an integrated simulation environment, we can accelerate VAS evaluation, getting closer to matching its fast-paced development and deployment of new tools².

In service of policymakers. We believe our work may help public servants such as regulators and national market authorities to decompose sources of impact of data-dependent algorithmic systems. For instance, our work helps organizations *locate their obligations* by decomposing a system into data, model, policy, and interface layers; VASs pinpoint where AI-critical artifacts (e.g., risk files) belong which are trackable by using various tools such as diagrams. Additionally, this helps decision makers to *quantify systemic risk* by connecting tool-based and system-based auditable metrics to societal outcomes, like fairness. *Unifying the language* of key concepts and relationships can help policymakers assess and audit risks of tools related to VASs, which hopefully contributes to developing harmonised standards. We hope our framework can be in service of regulators and national market authorities in their work by providing a new set of tools. We recommend policymakers to employ our proposed diagrams and documentation in their evaluation of impacts in VAS-related high-risk systems.

7 Conclusion

Our work introduces visibility allocation systems (VASs) and compatible definitions for their principal comprising tools. It also extends data flow diagrams to represent sociotechnical systems relying on VASs and to support the evaluation of their long-term effects. The result is a unified model that includes both search and

recommendation, together with their objectives, data transformation, and impacted stakeholders. Using school choice as an example, we demonstrate how our framework can support documenting findings and system design changes. Ultimately, the framework helps connect prior results from different research fields, including Information Retrieval and Recommender Systems, under one model and reason about the societal impact of design choices.

Altogether, we believe our work can support contemporary research and legislative efforts for increasing transparency, promoting understanding for market operators, and developing a harmonized standard for assessing and managing risks across stakeholders [50]. We view this work as a first step toward building an integrated simulation environment for VAS-based sociotechnical systems, where researchers and developers in different fields can independently contribute with their own implementation (e.g., for recommender systems [63] or human behavior in response to VAS-outputs [97, 118]) toward analyzing the risks of VASs and their alignment with human values. As proved by other domains [68, 86, 128], building such an environment could accelerate VAS evaluation, getting us closer to matching their fast-paced development.

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²To demonstrate the feasibility of doing so, we have developed a Python interface that enables users to define dataflow, tools, and metrics. We include this implementation in an anonymous GitHub repository: <https://anonymous.4open.science/r/VAS-anonymized-submission-D122/README.md>.

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